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The Economics of Nonmarital Childbearing and the Marriage Premium for Children

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Abstract

A large body of literature exists on the impact of family structure on children's outcomes, typically focusing on average effects. In this review, we build on this with an economic framework that has heterogeneous predictions regarding the potential benefit for children of married parents. We propose that the gains due to marriage from a child's perspective depend on a mother's own level of resources, the additional net resources that her partner brings, and the outcome-specific returns to resources. Data from the Panel Study of Income Dynamics are consistent with the heterogeneous predictions of this framework. In terms of high school completion or avoiding poverty at age 25, the so-called marriage premium for children is highest for children of mothers with high school degrees and mothers in their early to mid-20s. For the more advanced outcomes of college completion or high income at age 25, the marriage premium monotonically increases with observed maternal age and education.



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1. INTRODUCTION

For much of US history, a married-couple family constituted the primary unit of social and economic life for individuals. Married couples birthed and raised children, pooled resources of time and money, and provided for each other into old age. However, this relationship has been breaking down for decades, particularly in the context of childrearing.¹ By 2014, over 40% of births in the United States took place outside marriage. The rate is particularly high among certain population subgroups: 71% among African-Americans, 71% among women under age 25, and 62% among women with a high school degree or less education.² This reflects a dramatic shift over the past 50 years. In 1960, only 5% of children were born to unmarried mothers.

One reason that the high level of nonmarital childbearing is an economic issue is because numerous studies show that children raised in single-parent homes fare worse on a number of educational and economic dimensions as compared to children raised in homes with two married biological parents. Conditional on other family and parental characteristics, children who grow up with an unmarried mother have lower levels of educational attainment and higher levels of teen childbearing, among other outcome differences (e.g., Hoffman et al. 1993, McLanahan & Sandefur 1994, McLanahan 1997, Amato 2005). As adults, they have lower levels of employment, lower levels of income, higher rates of nonmarital parenting, and higher rates of incarceration (e.g., Hill et al. 2009, Lopoo & DeLeire 2014, Lerman et al. 2017).³ Although none of these studies completely surmounts the challenge inherent in identifying whether the differences are due to marriage per se, rather than associated unobservable resources or parental characteristics, the strength of the association and its robustness to a myriad of controls for observable factors is at least strongly suggestive of a causal relationship.⁴

The academic literature and policy conversation about nonmarital childbearing, however, generally do not distinguish among different instances of nonmarital childbearing. Having a child outside of marriage presumably has different implications for a 19-year-old woman without a high school degree than it does for a 25-year-old with some college education or a 35-year-old college graduate. To put it differently, the advantage that marriage would bring to these children is likely to be different across circumstances. Perhaps older and more educated unmarried mothers are able to provide sufficiently for their children alone such that their children do not suffer any sort of disadvantage in terms of long-term economic outcomes. Perhaps the men who father children

¹In this review, we do not explicitly consider the question of what is responsible for driving the rise in nonmarital childbearing. Ellwood & Jencks (2004a,b) provide an overview of the theoretical and empirical literature on the rise in nonmarital childbearing. They note that traditional economic models have highlighted four potential factors: (a) male earnings (e.g., Wilson & Neckerman 1986), (b) female earnings, (c) the sex ratio, and (d) public assistance (e.g., Moffitt 2001). Recent work by Shenhar (2015) explores the rise in female relative wages as an explanation for the decline in marriage rates. Ellwood & Jencks (2004a,b) summarize the main noneconomic explanations that have been advanced, including (a) gender role conflict, (b) limited confidence and personal efficacy, (c) altered attitudes and social norms, and (d) technological and legal change (e.g., Akerlof et al. 1996). Lundberg & Pollak (2007) provide a thorough review of how the US family has changed between 1960 and 2006.

²We calculated these statistics based on data available from the Centers for Disease Control WONDER online database, accessed June 20, 2016 (<https://wonder.cdc.gov/>).

³Recent work in economics has considered whether the incidence of single-mother households is related to racial and gender gaps in behavioral and educational outcomes. Hill et al. (2009) explore the extent to which racial differences in family structure might explain racial gaps in education, employment, and incarceration outcomes. Autor et al. (2015) and Bertrand & Pan (2013) suggest that the growing gender gap in behavioral and educational outcomes for children is at least partly explained by the relative disadvantage of boys who grow up in single-mother homes.

⁴Hill et al. (2009) and Lopoo & DeLeire (2014) provide an excellent review of the literature on family structure and children's outcomes.

with younger, less-educated women are of sufficiently low income and parenting quality that such a marriage would not appreciably improve a child's long-term economic prospects.⁵

In this review, we consider the notion of a marriage premium for children through the lens of economics, focusing on potential resources and returns. We introduce this term as a shorthand label for the difference in outcomes between children who are born to married parents and those born to a single mother. We maintain an empirical focus on the mother's marital status at birth mainly for analytical convenience, although a mother's marital status at birth is highly predictive of the family structure a child will experience during their childhood in the United States. We define resources to be those relevant to child development, including parental income, wealth, and time.⁶

The documented advantage, on average, for children from two-parent homes is consistent with a model where resources are higher, on average, in married, two-parent households. Not only does having two married parents in the household increase the total level of resources by virtue of having two resident parents present, this family structure capitalizes on economies of scale in a household and enables spouses to specialize across market work, home production, and child rearing, as in Becker's (1981) seminal model. A straightforward extension of this reasoning indicates that the relative advantage to a child of having married parents will vary with maternal circumstances in systematic ways. To be sure, the observed advantage for children is also consistent with a positive selection story: Conditional on observables, individuals who have children within marriage might possess inherent, unobservable qualities that benefit child development. Although we do not discount the role of selection, we are interested in developing a resource-based model that can make predictions about when the benefits of a married household may be particularly large from the perspective of the child.

We propose that the gains due to marriage from the perspective of a child will depend on (a) the mother's own level of resources, (b) the additional resources that her marriage would bring to the household, and (c) the returns to those additional resources. The resource gain will depend on the nature of marriage markets and the resources of the mother's partner. The return to resources will depend on the nature of the particular production function for outcomes, which will vary across outcome type. This set of factors and the interactions among them will generate heterogeneous marriage premiums for children born to mothers with different characteristics and for different types of outcomes. For example, for some women, a partner's additional contributions are not sufficient to alter children's outcomes, whereas for other women, a partner's contributions are not necessary. For instance, even if a teen mother married the father of her child, their combined resources might not be sufficient to equip their child to attain higher levels of education. In contrast, a highly educated older mother might have sufficient resources as a single mother such that her child suffers no relative disadvantage when it comes to educational attainment.

We explore these relationships empirically, motivated by our proposed marriage premium framework. We rely mainly on data from the Panel Study of Income Dynamics (PSID). This data

⁵Thomas & Sawhill (2002) offer evidence to the contrary; their simulations find that marriages between single mothers and men with similar characteristics would appreciably reduce child poverty rates.

⁶There are a variety of mechanisms beyond income through which children might benefit from living in a married-parent home. Many academic articles have reviewed potential mechanisms, including an overview piece by Ribar (2015). Studies that have attempted to investigate how much of the disadvantage observed for children from single-parent homes is due specifically to lower household income generally find that the direct effect of income is an important part of the story, but not the full explanation, though this remains an open question (see, for example, McLanahan & Sandefur 1994, Hill et al. 2009, Lopoo & DeLeire 2014, Lerman et al. 2017). Low levels of income and single parenthood are often associated with other stress factors that are directly harmful to children's development, such as residential instability, maternal stress, and less positive parenting practices (see also Mayer 1997a,b,c; Kalil et al. 1998; Furstenberg et al. 1999).

source allows us to observe adult outcomes for children born in the 1960s, 1970s, and 1980s. We consider two types of long-term outcomes for children. First, we consider primary outcomes, such as graduating from high school and having income above the federal poverty threshold at age 25, which are commonly achieved. Second, we consider advanced outcomes, such as graduating from college and having high income (above 400% of the federal poverty threshold) at age 25, which are achieved by fewer people. We use these data to estimate descriptive relationships between these outcomes and mother's age at time of child's birth and mother's level of education, with the goal of documenting patterns of the marriage premium for children across the mother's characteristics and outcomes. We do not attempt to empirically identify a causal relationship between marriage as a family structure and children's outcomes; rather, we explore systematic patterns of marital gaps in children's outcomes by baseline maternal characteristics.

The patterns we observe for being out of poverty by age 25 and graduating from high school suggest that the marriage premium has an inverted U-shaped pattern along the dimensions of maternal age and education. For the youngest and least-educated mothers, as well as for the oldest and most-educated mothers, there does not appear to be a large marriage premium for children. The greatest marriage premium occurs in the middle of the maternal age and education distributions. This is consistent with the predictions of our model given assortative mating and reasonable expectations about the return to resources for these primary outcomes. The potential resource gain from marriage is not sufficient at the low end, and is unnecessary at the high end, to substantially increase a child's probability of avoiding poverty or completing high school. Marriage is associated with the largest differences in these two outcomes for mothers in their early to mid-20s and those with a high school degree.

For the advanced outcomes of graduating from college and having a high level of income at age 25, the observed marriage premium increases monotonically with maternal age and education. This is also consistent with the predictions of our model given assortative mating and reasonable expectations about the return to resources for an advanced outcome. In the case of these harder-to-obtain outcomes, it appears that marriage is associated with an increasingly higher likelihood of achieving these advanced outcomes for children up to the highest levels of observed maternal age and education. In a final empirical consideration, we consider the extent to which differences in household income explain the observed premium for both types of outcomes. The data suggest that household income at birth is only partly responsible for these gaps.

2. THE MARRIAGE PREMIUM FOR CHILDREN: A CONCEPTUAL FRAMEWORK

In this section, we present a conceptual framework for thinking about a marriage premium for children as a function of underlying resources and returns to resources. To define terms and notation, let us use α to denote the difference in outcomes between children raised in a household headed by married parents and those raised in a household headed by a single mother. Specifically, we propose that the marriage premium for children (α) will depend crucially on three elements: (a) the mother's own level of resources, (b) the additional resources that marriage would bring to the household, and (c) the returns to those additional resources.

We denote a mother's and father's own level of resources as R_m and R_f , respectively, and total household resources as R_t : $R_t = R_m + R_f$. We use the term resources broadly to signify the range of positive inputs that parents bring to a child's experience. This includes not only financial resources and parental time, but also additional elements such as social and professional networks, positive parenting practices, physical and mental well-being, and home life stability. For unmarried mothers, we impose an expositional simplification such that total household resources are fully

captured by mother's own resources ($R_t = R_m$), which is inclusive of any resources she obtains on behalf of her child, whether through her own earnings or time, through government support, through her network of family and friends, etc. If the mother were to marry the father of the child and co-reside, household resources would increase by the net level of resources the father brings to the household (beyond what he would contribute as a nonresident, unmarried father), R_f .⁷

2.1. Assortative Mating

The potential resource gain that marrying the father of a child would bring to the household depends on the father's level of resources. That gain will depend on the nature of the market for sexual partners and marriage (which we take to be the same). In the United States, there is a high degree of assortative mating such that individuals tend to partner with and marry individuals of a similar socioeconomic status (SES) (see Schwartz 2010). We incorporate this into our model by treating resource level as isomorphic to SES and assuming assortative mating based on relative resource position.

In a highly stylized version of assortative mating, women and men match based on their percentile in some underlying resource distribution. If the underlying male and female resource distributions were equal, then assortative mating of the type we describe would result in a doubling of potential household resources if the parents were to marry. That is, R_m at the 25th percentile of the distribution among mothers plus R_f at the 25th percentile of the distribution among fathers would double R_t . These observations lead to the following proposition relevant for thinking about how marriage would heterogeneously increase resources available to a child:

Proposition 1: Given assortative mating, the absolute resource gain from marriage ($R_t - R_m = R_f$) increases in mother's own resources (R_m): $d(R_t - R_m)/dR_m > 0$.

2.2. Production Functions for Children's Outcomes

The production function for child outcomes dictates how beneficial the additional resources that a married father brings to the household are for a child's development and, ultimately, adult outcomes. Let us assume that the production of positive child outcomes (e.g., educational attainment) responds positively to parental resources such that higher levels of R_t (as inputs) produce increased levels of child educational and socioeconomic achievement (as outcomes or output).⁸

⁷We abstract from the situation of cohabitation among unmarried parents in the model. As a practical matter, in the United States, the situation for children of cohabiting parents tends to fall somewhere in between the more common situations of unmarried and noncohabiting parents and married parents, but it is generally understood to involve much higher rates of family instability compared to married families. Manning (2015) reviews the issue of cohabitation and child well-being. She notes that more children born to cohabiting parents see their parents break up by age 5, as compared to children born to married parents. Only one out of three children born to cohabiting parents remains in a stable family through age 12, in contrast to nearly three out of four children born to married parents. She further reports that children born to cohabiting parents experience nearly three times as many family transitions as those born to married parents (1.4 versus 0.5).

⁸Several authors have attempted to produce credible estimates of the causal relationship between family income and child outcomes using exogenous variation in family income (see, for example, Milligan & Stabile 2007, Oreopoulos et al. 2008, Dahl & Lochner 2011, Løken 2010). Løken et al. (2012) estimate nonlinear instrumental variables and fixed effects models relating family income to child outcomes. Their analysis uses administrative registry data for the entire population of Norway, with information on children's educational attainment and IQ as adults as well as their family income during childhood. They instrument for family income with regional and time variation in the initial discovery of oil in Norway. They find an increasing, concave relationship between family income and children's outcomes. They observe that previous studies based on linear estimators understate the beneficial effect of family income for children because they assign little weight to the large marginal effects at lower levels of income. This econometric point is related to, albeit distinct from, the conceptual point we make in Section 3 about heterogeneity in the returns to additional parental resources.

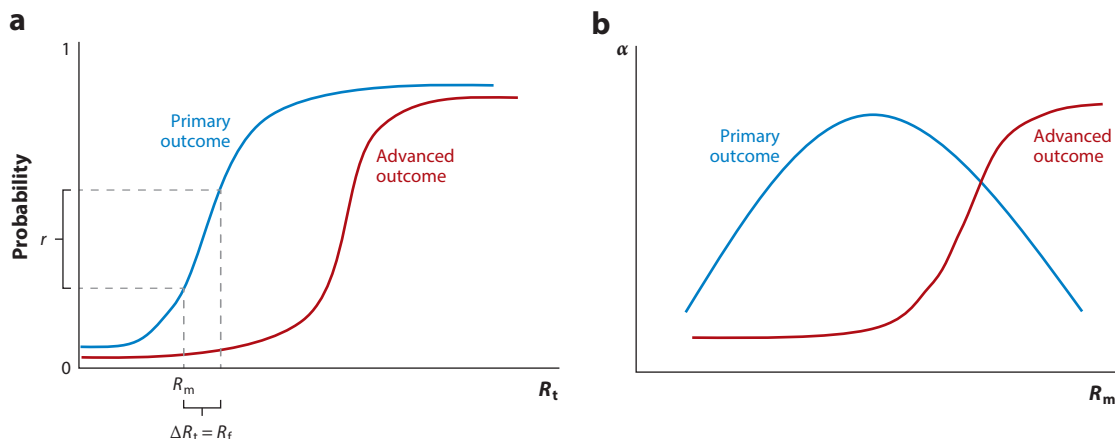


Figure 1

Deriving the marriage premium for children. (a) Hypothetical impact of total household resources (R_t) on probability of achieving primary and advanced outcomes. R_m and R_f represent the independent level of resources of the mother and father, respectively. The return to the resource gain that comes from marriage is given by r . (b) Marriage premium for children. The difference in outcomes between children raised in a household headed by married parents and those raised in a household headed by a single mother is given by α .

For simplicity, we introduce the notion of two types of outcomes: primary and advanced outcomes. Primary outcomes consist of those that one might reasonably consider first-order markers of economic success. The majority of individuals would achieve these: progressing through grade levels at school appropriate for their age, avoiding a teen birth, completing high school, and living above the poverty line as an adult. Advanced outcomes consist of outcomes achieved by a smaller minority of individuals. Such outcomes would include gaining admittance to a selective college, completing a college degree, and obtaining a high-paying job.

The return r to the resource gain that comes from marriage (through the addition of R_f to the household) will lead to an improvement in child outcomes y as determined by the relationship between the outcome and the overall resource level. Let us call this production function q . **Figure 1a** plots the relationship between total household resources R_t and a primary (p) and advanced (a) outcome. Although we assume simple functional forms q_p and q_a for illustrative purposes, the main feature of the figure is that the production function (expressed as a probability function) for the advanced outcome falls to the right of the probability function for the primary outcome. As a father's additional resources are brought into a household headed by a mother with fewer resources (a younger or less educated mother), the probability that the child achieves a primary outcome is increased. The total resource level, however, is still too low for an appreciable change in the likelihood of achieving the advanced outcome. The additional resources from that marriage would have to be extensive to have an impact on an advanced outcome.

In contrast, when a father's additional resources are brought into a household headed by a mother with more resources (for instance, an older or more educated mother), primary outcomes are unlikely to be affected. The mother herself is likely to have sufficient resources to enable her children to achieve that outcome. Her children would benefit by becoming more likely to achieve an advanced outcome if marriage occurs. These observations lead to the following propositions:

Proposition 2: For primary outcomes, the children of lower-resourced mothers will experience the largest relative benefit from marriage as long as fathers have non-negligible resources to contribute to the household.

Proposition 3: For advanced outcomes, the children of higher-resourced mothers will experience the largest relative benefit from marriage.

2.3. The Marriage Premium for Children

Building on the notation introduced so far and combining the insights from Propositions 1, 2, and 3, we can write an expression for the marriage premium to children as follows:

$$\alpha = r_y * (R_t - R_m) | R_m.$$

This expression indicates that the marriage premium α is a function of the return to resources for a given outcome (r_y) interacted with the difference in total household resources that would result from marriage ($R_t - R_m$), conditional on the baseline level of resources a mother brings to the household in the nonmarital situation.

We have presented this model in terms of a generic resource that is essentially isomorphic to income and increases in maternal age and education. An obvious extension of the model would be to consider different types of resources. For example, one could model the marriage premium as a function of separate resources such as income, parental time, and family stability and make assumptions about whether these resources operate as complements or substitutes. Another straightforward adaptation would be to specify the marriage premium as being net of certain resources, for example, a marriage premium net of income effects. We offer the framework in this section as a basic model that can be readily extended and adapted for more nuanced considerations of the relationship between parents' marital status and children's outcomes.

3. MARRIAGE GAPS IN MATERNAL CHARACTERISTICS AND BASELINE RESOURCE LEVELS

The discussion in the previous section emphasizes the systematic ways we would expect the marriage premium for children to vary with maternal characteristics and outcome types. Before exploring the heterogeneity in outcomes directly, we provide a preliminary, descriptive look at the data. We first confirm that maternal marital status at the time of a child's birth is highly predictive of a child's family structure at age 14. We then document average marriage gaps in maternal characteristics, household resources, and children's outcomes and how they have changed over time.

In terms of maternal characteristics, we focus on mother's age and education, with the presumption that older, more educated mothers offer higher levels of parental resources (for evidence in support of this presumption, see Miller 2009). As supporting evidence, we calculate using the 2014 American Community Survey that, among women with children age 5 or under who work more than 1,500 hours per year, those age 35 and over have median annual earnings of \$50,000 in 2013 compared to \$18,000 for those under age 25. Similarly, college-educated mothers in this group earn 2.4 times as much as those with a high school degree, \$55,000 relative to \$23,000. In addition, research has documented that more educated mothers spend more time with their children (Guryan et al. 2008) and are more likely to alter the composition of that time to suit children's developmental needs (Kalil et al. 2012).

Table 1 Family structure transition rates between birth and age 14

Family structure at age 14	Family structure at birth: born 1960–1979			Family structure at birth: born 1980–1999		
	Married parents	Cohabiting parents	Mother only	Married parents	Cohabiting parents	Mother only
Married parents	80.5	39.5	15.1	74.9	27.9	11.6
Cohabiting parents	0.4	17.4	2.9	0.6	10.9	1.2
Mother and stepfather	6.7	3.1	12.1	7.1	9.1	9.1
Mother only	9.0	16.3	62.9	12.0	33.3	64.9
Other	3.4	23.7	6.9	5.4	18.8	13.2
Percent in category	82.8	2.0	15.1	76.1	7.7	15.9

An observation is a child born in the stated years in the Panel Study of Income Dynamics (PSID). Family structure at birth is based on data from the first survey after the child was born. “Mother only” includes living with the mother and grandparents. “Other” includes living with the father only or with neither parent. Each observation is weighted by the mother’s individual-level PSID sampling weight for personal characteristics and by the household PSID sampling weight for household-level measures.

We rely primarily on data from the PSID for this and subsequent analyses. The PSID is a nationally representative, family-based dataset that first sampled over 18,000 respondents in 1968. Since then, the survey has followed all of the original participants and all of those who joined their families, including children. The most recent data available at the time of this analysis is from 2013. In these data, we observe maternal characteristics at the time of birth (marital status, age, education, and income) and subsequent outcomes for the children (poverty status, high school completion, and college completion). Because we are interested in observing children’s long-term outcomes, we examine the experiences of children born during the 1960s, 1970s, and 1980s. An online **Supplemental Appendix** (follow the **Supplemental Materials** link at <http://www.annualreviews.org/>) details how we construct our analysis sample and variables from the PSID. We augment our PSID analysis with additional data on births from the 1980 and 2014 Vital Statistics natality files to document characteristics of mothers for two cohorts and observe how maternal characteristics have changed over time. The issue of drawing on lessons from earlier cohorts to inform the experience of today’s children is something we return to in our final discussion (Section 5).

3.1. Transitions in Family Structure

In our analysis, we focus on the mother’s marital status at the time of childbearing. This coincides with our emphasis on the impact of nonmarital childbearing, but it is also a strong predictor of subsequent living arrangements for children. This can be seen in **Table 1**, where we examine children’s living circumstances at age 14 and their mother’s marital status at their birth. We examine children born between 1960 and 1979 separately from those born between 1980 and 1999 (who were 14 years old in 2013, the last year of data available) to gauge the extent of changes over time. Consistent with trends documented elsewhere, the bottom row indicates that fewer children were born to married mothers in the more recent period: 76%, down from 83%. It also shows that births to cohabiting parents rose over that interval, from 2% in the earlier period to 7.7% of births in the more recent cohort.⁹

⁹Manning (2015) reports that nearly one-quarter of children born today are born to cohabiting parents.

These data indicate that initial family structure at birth is strongly persistent to age 14. Among children born between 1980 and 1999, 75% of those born to married mothers are still living in a household composed of married parents, and 65% of those born to a single mother are living in a mother-only household, at age 14. Only 12% of children born to single mothers see their parents married at age 14. Being born to cohabiting parents is a less persistent family structure, as has been well documented elsewhere (see, for example, Manning 2015). Among children born to cohabiting parents (7.7% of children in the more recent cohort), 28% are living with married parents, 33% are living with their mother only, and 9% are living with their mother and stepfather at age 14.

3.2. Mother's Characteristics at the Time of Birth

Our next set of analyses uses Vital Statistics natality data to document age and education distributions of marital and nonmarital births. These analyses are motivated by the notion that older, more educated mothers are likely to have higher levels of resources as compared with younger, less educated mothers. We look separately at births occurring in 1980 and in 2014. The Vital Statistics data include virtually all births in the United States. **Figure 2** reveals clearly that the age distribution of married mothers skews to the right (toward older ages) as compared to the age distribution of unmarried mothers. In 2014, the modal age of a married mother was 31, compared to 22 for unmarried births. Mothers were younger in 1980, with the modal age being 25 among married mothers and 19 among unmarried mothers. The data also reveal a reduction, over time, in teen childbearing and an increase in childbearing during ages 30–44 among all mothers.

To the extent that older and more educated parents bring greater resources to a household, the older ages of mothers in more recent years would imply improved outcomes for all children. However, the gap in mother's modal age between marital and nonmarital births has actually jumped from 6 years to 9 years. As much as the universal increase in mother's age could improve outcomes for all, it could also increase the marriage premium for children (if greater maternal age implies higher levels of parenting resources).

We observe a similar pattern in **Figure 3**, which plots the education distribution of mothers by marital status. Married mothers tend to be more highly educated than mothers who give birth outside of marriage. In 2014, the modal married mother was a college graduate: 45.8% of married mothers had a college degree or higher. Only 7.2% of mothers who were unmarried at the time of birth had a college degree or higher. The modal unmarried mother at the time of birth was a high school graduate (36%). Only 8.5% of married mothers had less than a high school degree, as compared to 25% of unmarried mothers. As shown in **Figure 3a**, the 1980 education distributions of both sets of mothers are shifted to the left (toward lower levels of education), with many fewer mothers having a college degree at the time of giving birth and many more having less than a high school degree. Interestingly, the gains in education have been greater among married mothers than among unmarried mothers, so that, even though all mothers were more educated in the more recent cohort, the increase in education between 1980 and 2014 among married mothers was greater than that among unmarried mothers.

Table 2 reports similar statistics using PSID data, breaking up the sample of children into those born to married and to unmarried mothers in the 1968–1989 and 1990–2013 windows. Again, education and maternal age are greater for married mothers and rising over time for both married and unmarried mothers, but rising more for married mothers. The data also reveal that a married mother is extremely unlikely to live with her own mother, but it is reasonably common

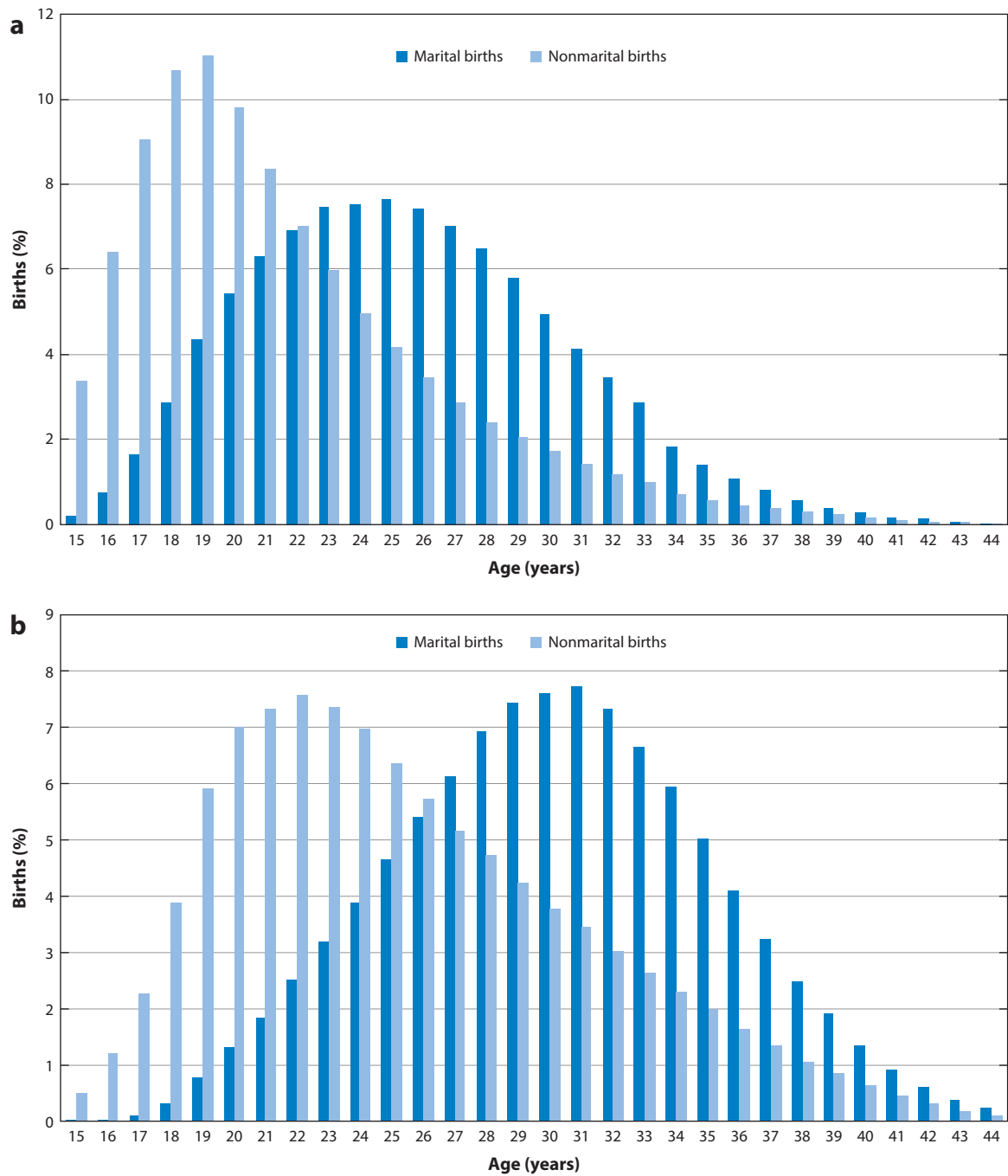


Figure 2
Distribution of mother's age at birth of child, by mother's marital status, in (a) 1980 and (b) 2014. Figure created with the authors' calculations using Vital Statistics natality microdata.

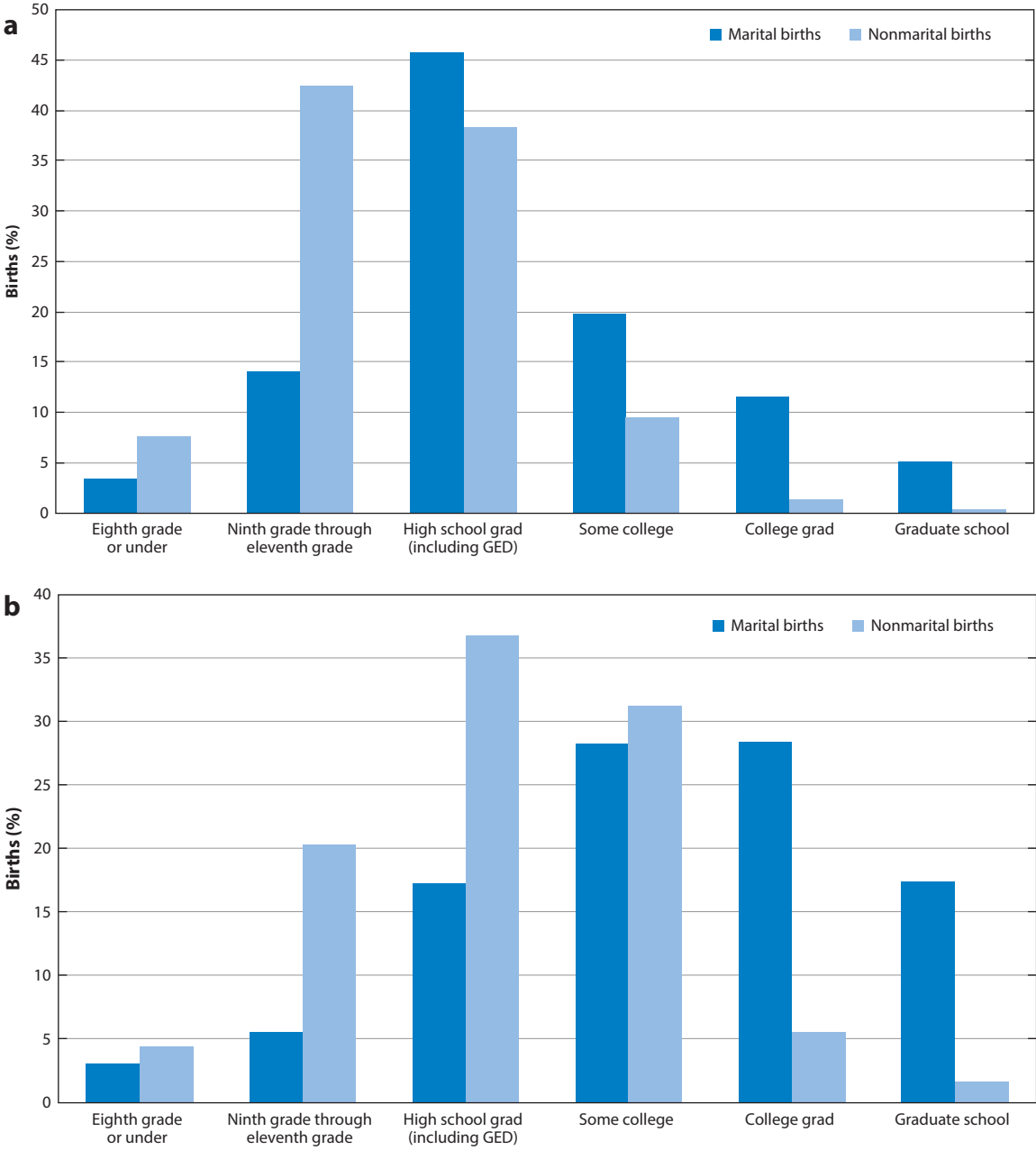


Figure 3
Distribution of mother's educational attainment at birth of child, by mother's marital status, in (a) 1980 and (b) 2014. Figure created with the authors' calculations using Vital Statistics natality microdata.

Table 2 Mother and household characteristics in year after birth

	Births, 1968–1989			Births, 1990–2013		
	Married	Not married	Gap (married–not married)	Married	Not married	Gap (married–not married)
Mother's characteristics						
Age at child's birth	26.6	22.8	3.8	29.8	23.7	6.1
Years of education	13.5	12.4	1.1	14.6	12.8	1.8
Percent living with parents	1.7	32.0	–30.3	2.4	28.2	–25.8
Household income at birth						
Median income	\$50,222	\$21,774	\$28,447	\$73,255	\$31,329	\$41,926
Percent not in poverty	90.8	48.9	41.8	91.0	61.1	29.9
Household income in childhood						
Median income, ages 0–5	\$54,677	\$24,561	\$30,116	\$77,090	\$33,406	\$43,684
Percent of years not in poverty, ages 0–5	90.6	54.0	36.5	91.1	62.7	28.5
Median income, ages 0–16	\$64,651	\$28,848	\$35,803	\$82,454	\$35,430	\$47,024
Percent of years not in poverty, ages 0–16	90.9	57.2	33.7	91.4	63.6	27.8
Observations	6,126	2,232	—	4,983	3,620	—

An observation is a child born in the stated years in the Panel Study of Income Dynamics (PSID). Each observation is weighted by the mother's individual-level PSID sampling weight for personal characteristics and by the household PSID sampling weight for household-level measures. Household income is reported in 2013 dollars, and adjustments have been made using the personal consumption expenditures price index. Income between ages 0–16 is the child's annual household income averaged over all available years between the ages of 0 and 16, with the median reported.

for an unmarried mother to do so. This could reflect her greater need for support but also the greater resources available for her children than if she lived alone.

3.3. The Resource Gap at the Time of Birth

Table 2 also provides summary statistics of family income as a direct measure of the resource gap at the time of birth. From these statistics, we see that household income at birth is greater for children born to married mothers and the gap in absolute terms increases over time. For the cohort born between 1990 and 2013, median household income (in year 2013 dollars) was \$73,255 for married mothers and \$31,329 for unmarried mothers, a difference of \$41,926. For the earlier cohort, the income advantage of married households was \$28,447. The data also show that, in both periods, many more children born to unmarried mothers lived in poverty at the time of birth compared to children born to married mothers: roughly 40% of the later cohort and more than 50% of the earlier cohort. In both periods, less than 10% of children born to married parents lived in poverty at the time of birth.

We also examine differences in the percent of childhood (through age 16) spent in poverty. We distinguish children by birth cohorts 1968–1989 and 1990–1997 and restrict the sample to those children who have reached age 16 by the 2013 date of the last available survey. For both cohorts, children born to married parents, on average, spent just over 90% of their childhood years out of poverty. In contrast, for both cohorts, children born to an unmarried mother spent around 60% of their childhood above the poverty line.

Data from the American Time Use Survey from 2003 to 2014 indicate a resource gap between children born to married and to unmarried parents in terms of parental hours spent per week on childcare.¹⁰ We look at parents' reported time with children and focus on children under age 5. Married mothers spend 18 hours per week caring for their young children compared to 14.9 hours per week for unmarried mothers. Married fathers also spend an additional 1.4 hours per week, on average, caring for their children (9.8 hours per week compared to 8.4 hours per week). These gaps are not measured at the time of birth, as our preceding measures of resources have been, but the correlations over time for preschool age children are likely to be very high. These average differences reveal that young children of married parents generally spend more time with their parents compared to children of unmarried parents.¹¹

3.4. Outcome Gaps Associated with Maternal Marital Status

Consistent with the documented fact that children born to unmarried mothers have fewer resources in their households, the data also indicate that these children experience inferior educational and labor market outcomes relative to those children born to married mothers. **Table 3** displays these patterns and confirms those reported in previous studies (cited above). In our analysis, we examine four specific outcomes: being out of poverty at age 25, completing high school by age 20,¹² having household income over 400% of the poverty level at age 25 (to indicate high income), and completing college by age 25. We separate children into cohorts by birth decade to examine trends over time. In all cases, outcomes of children born to married parents are better in these dimensions than those of children with unmarried parents at birth.

One notable pattern in these data is that recent birth cohorts show a larger marriage premium in the likelihood of college completion compared to earlier cohorts. The likelihood that children born to unmarried mothers will complete college increased from 5.9% for children born in the 1960s to 11.3% for children born in the 1980s. The likelihood for children born to married mothers, however, increased from 21.8% to 40.7%, resulting in an approximate doubling of the gap between the two groups (from 15.9 percentage points to 29.4). However, the marriage premium in rates of high school graduation fell slightly. The gap likelihood of being above the poverty threshold rose and then fell, ending up back at its original level.

These patterns are consistent with the predictions of our conceptual model of a heterogeneous marriage premium. Rising out of poverty and graduating from high school are primary outcomes within that framework, whereas having a high income and graduating college are advanced outcomes. Increases in parental resources over time are only going to have an impact if a substantial share of some group is below that threshold and is lifted over the bar because of the additional resources brought to the household through marriage. For a primary outcome, a young single

¹⁰Time spent caring for children is defined as the sum of time spent by parents on basic, educational, and recreational childcare. The full sample includes all individuals between the ages of 18 and 55 who report at least one child under the age of 5. The resulting sample includes 8,615 mothers and 6,093 fathers. Means are calculated using fixed demographic weights adjusted to represent each day of the week equally within subgroups.

¹¹This finding is consistent with the results of a study by Kalil et al. (2014) that uses the 1997 PSID Child Development Supplement to examine time investment in children across six family structures. The authors document that children receive more total caregiving time in married-biological-parent families than in single-mother households because those households receive little time investment from their nonresident biological fathers.

¹²The PSID data does not allow us to consistently separate GED recipients from traditional high school graduates. As such, this measure captures high school graduation or its equivalent.

Table 3 Children's long-term outcomes by child's year of birth and mother's marital status at birth of child

Child's year of birth	Married	Not married	Gap (married–not married)	Number of observations
Percent not in poverty at age 25				
All sample years	91.6	72.6	19.0	6,444
Born 1960–1969	91.0	74.0	17.0	2,288
Born 1970–1979	93.3	69.6	23.7	2,016
Born 1980–1989	90.6	74.1	16.5	2,140
Percent with a high school degree by age 20				
All sample years	88.1	78.1	10.0	8,225
Born 1960–1969	86.9	74.7	12.2	2,610
Born 1970–1979	89.0	78.2	10.8	2,502
Born 1980–1989	88.4	78.8	9.6	3,113
Percent over 400% of the poverty level at age 25				
All sample years	39.7	15.9	23.9	6,444
Born 1960–1969	38.4	17.1	21.3	2,288
Born 1970–1979	41.4	12.6	28.8	2,016
Born 1980–1989	39.5	17.5	22.0	2,140
Percent with a college degree by age 25				
All sample years	29.9	9.5	20.4	6,847
Born 1960–1969	21.8	5.9	15.9	2,307
Born 1970–1979	28.0	8.1	19.9	2,095
Born 1980–1989	40.7	11.3	29.4	2,445

An observation is a child born in the stated years in the Panel Study of Income Dynamics (PSID). Each observation is weighted by the child's individual level PSID sampling weight in the year in which the outcome was observed.

mother might not have sufficient resources to clear that bar on her own, but marriage has the potential to bring additional resources that would raise children over the poverty threshold (see Thomas & Sawhill 2002). Older, more educated mothers are likely to have enough resources to avoid poverty even without the additional resources of a spouse.

For an advanced outcome, however, combined resources from marriage may potentially increase total household resources by a large amount for a highly educated mother, given that the father of her child is likely to be a high-resource man (through assortative mating). College completion rates increase with family income.¹³ It therefore makes sense that marriages associated with a greater base level of resources from the mother and relatively large resource gains from the father will be associated with increasingly high rates of college attendance, and that children from lower-resourced households will fall further behind.

¹³Bjorklund-Young (2016) reports that data from the Education Longitudinal Survey—a nationally representative sample of students who were high school sophomores in 2002—indicate that, among students from the lowest quartile of family income, college completion rates are 14%, compared to 60% among students from the top quartile of family income.

Table 4 Children's long-term outcomes by mother's marital status and age at birth of child

Mother's age	Married	Not married	Gap (married–not married)	Number of observations
Percent not in poverty at age 25				
Mother <20 at birth	86.9	71.7	15.2	907
Mother 20–24 at birth	91.7	69.1	22.6	2,127
Mother 25–29 at birth	93.4	74.9	18.5	1,794
Mother 30+ at birth	91.2	75.1	16.1	1,533
Percent with a high school degree by age 20				
Mother <20 at birth	81.0	74.1	6.9	1,174
Mother 20–24 at birth	87.6	77.3	10.3	2,667
Mother 25–29 at birth	89.7	81.8	7.9	2,286
Mother 30+ at birth	88.7	80.2	8.5	1,981
Percent over 400% of the poverty level at age 25				
Mother <20 at birth	24.1	12.6	11.5	907
Mother 20–24 at birth	32.8	15.0	17.8	2,127
Mother 25–29 at birth	46.2	22.3	23.9	1,794
Mother 30+ at birth	43.9	19.0	24.9	1,533
Percent with a college degree by age 25				
Mother <20 at birth	13.1	6.6	6.5	954
Mother 20–24 at birth	21.9	5.8	16.1	2,241
Mother 25–29 at birth	34.5	15	19.5	1,914
Mother 30+ at birth	37.2	16.5	20.7	1,645

An observation is a child born between 1960 and 1989 in the Panel Study of Income Dynamics (PSID). Each observation is weighted by the child's individual-level PSID sampling weight in the year in which the outcome was observed.

4. EXPLORING HETEROGENEITY

Our model also makes predictions regarding the impact of marriage on mothers with different levels of resources. For primary outcomes, we hypothesize that the marriage premium for children has the shape of an inverted U, at first increasing and then decreasing as the mother's resources increase. For advanced outcomes, we anticipate that a marriage premium for children will exist at higher maternal resource levels. In this section, we report the results of an analysis designed to explore whether descriptive data are consistent with those hypotheses. As we have done throughout this review, we focus on maternal age and education as indicative of maternal resource level.

4.1. Descriptive Relationship Between Maternal Marital Status and Children's Outcomes

In **Tables 4** and **5**, we report the results of an analysis focusing on the same children's outcomes as reported in **Table 3**, except in this case we distinguish mothers by their age and educational attainment at the time of childbearing. All statistics reported are relevant sample means computed from the PSID. We do not attempt to isolate causal findings; instead, we explore heterogeneity in raw marriage premiums.

Table 5 Children's long-term outcomes by mother's marital status and mother's level of education

Mother's level of education	Married	Not married	Gap (married–not married)	Number of observations
Percent not in poverty at age 25				
Less than high school education	81.1	64.8	16.3	1,143
High school degree	90.9	73.7	17.2	2,499
Some college	93.8	76.0	17.9	1,629
College degree	95.7	80.5	15.2	1,162
Percent with a high school degree by age 20				
Less than high school education	73.6	67.9	5.6	1,485
High school degree	87.7	78.5	9.3	3,140
Some college	90.7	82.9	7.7	2,145
College degree	93.0	88.8	4.2	1,434
Percent over 400% of the poverty level at age 25				
Less than high school education	15.5	8.6	6.9	1,143
High school degree	37.4	16.5	20.9	2,499
Some college	40.9	17.7	23.2	1,629
College degree	54.1	29.1	25.0	1,162
Percent with a college degree by 25				
Less than high school education	7.4	4.9	2.5	1,212
High school degree	18.0	4.8	13.2	2,620
Some college	31.1	13.8	17.3	1,745
College degree	57.0	28.6	28.4	1,259

An observation is a child born between 1960 and 1989 in the Panel Study of Income Dynamics (PSID). Each observation is weighted by the child's individual-level PSID sampling weight in the year in which the outcome was observed.

The first thing to notice in **Tables 4** and **5** is that, for all four outcomes, there is a marriage premium observed for all maternal age and education categories. This is noteworthy given that one might have suspected older and more educated mothers would have sufficient resources on their own such that the children of married mothers would not experience an advantage in outcomes relative to the children of unmarried women. The data indicate that this is not the case. This result is consistent with previous studies that report the results of regression analyses that control for maternal age and education and still find disadvantages associated with single motherhood.

The patterns in the data are also consistent with the predictions of the model. For the primary outcomes of being out of poverty and graduating from high school, we see the inverted U-shaped pattern of the marriage premium for children. **Table 4** shows that women between the ages of 20 and 24 have children who receive the greatest observed marriage premium for these outcomes. That premium is lower for children of younger and older mothers. **Table 5** shows that women with a high school degree or some college education have the greatest observed marriage premium in terms of avoiding adult poverty and graduating from high school. Children of high school dropouts and college graduates have lower premiums, albeit still positive.

We see a different pattern when it comes to the advanced outcomes of college completion and having income above 400% of the federal poverty threshold (our measure of high income). In **Table 4**, we see that the marriage premium for advanced outcomes increases with maternal age. The children of married mothers age 30 and over are 20.7 percentage points more likely to

graduate from college than the children of unmarried mothers in that age group (37.2% compared to 16.5%). There is a positive age gradient in rates of college graduation for children of married and unmarried mothers, but the gradient is steeper for the children of married mothers. Among the children of unmarried mothers, the percent who complete college by age 25 increases from 6.6% for the children of teen mothers to 16.5% for the children of mothers age 30 and over. Among the children of married mothers, the percent who complete college by age 25 increases from 13.1% to 37.2%. A similar pattern holds for the advanced outcomes.

Table 5 reveals a similar story: The marriage premium also increases with maternal education. The children of married mothers with a college degree are 28.4 percentage points more likely to graduate from college than the children of unmarried mothers with a college degree (57% versus 28.6%). This gap is consistent with the predictions of our model, although we readily acknowledge the possibility that this result might, in part, reflect unobserved selection, meaning that unmarried, college-educated mothers might be more negatively selected than less educated unmarried mothers on other resource dimensions. As with maternal age, there is a positive education gradient for both sets of mothers such that higher levels of maternal education are associated with higher rates of college graduation among children, but the gradient is steeper for the children of married mothers. Among the children of unmarried mothers, the percent who complete college by age 25 increases from 4.9% for the children of mothers with only a high school degree to 28.6% for the children of mothers with a college degree, a nearly sixfold increase. Among the children of married mothers, the percent who complete college by age 25 increases from 7.4% to 57%, a nearly eightfold increase.

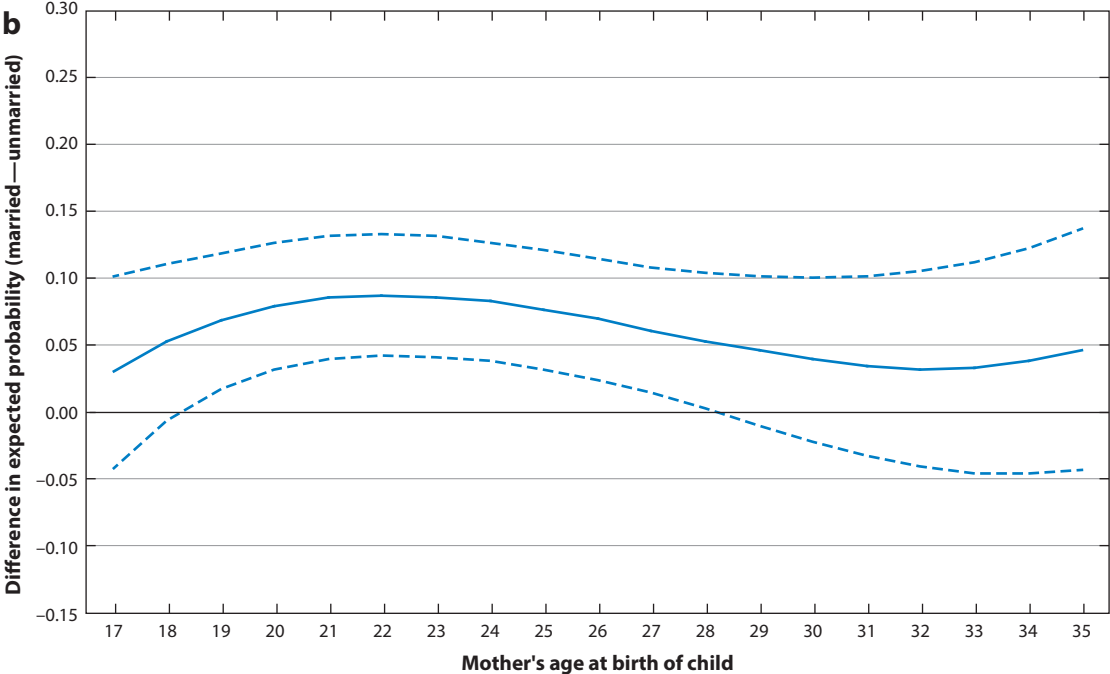
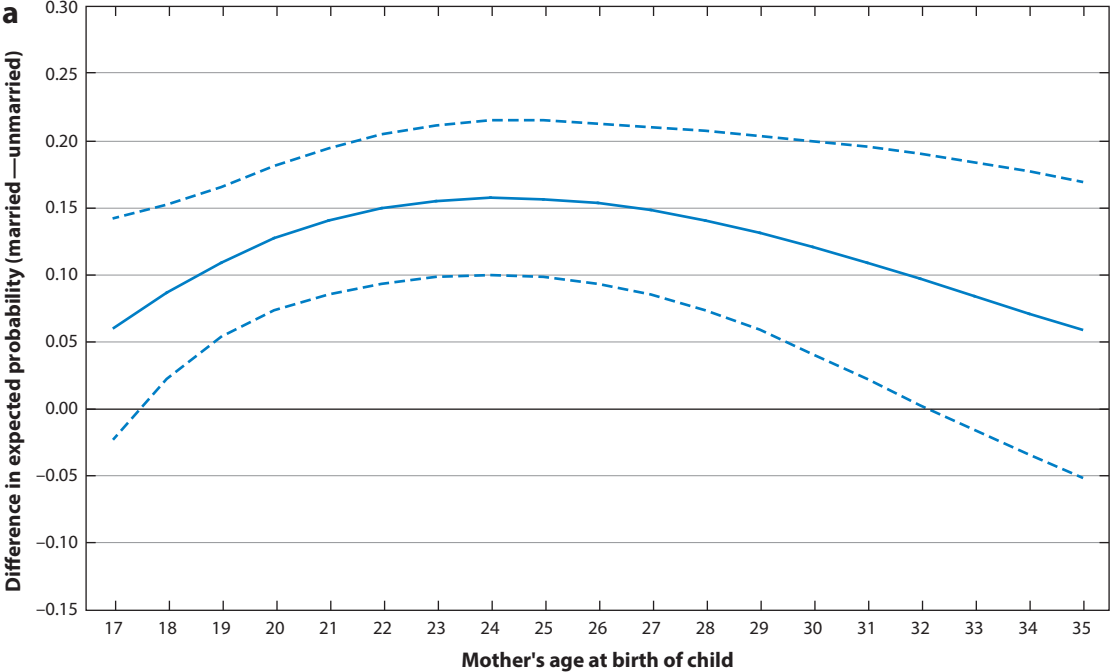
4.2. Multivariate Relationship Between Maternal Marital Status and Children's Outcomes

We extend this analysis by estimating regression models that parameterize these statistics and control for the impact that other demographic factors might have in the relationship between mother's marital status and children's outcomes. Specifically, we estimate models where the dependent variables are the same poverty and educational outcomes we have been using throughout this analysis. The key explanatory variables are interaction terms between maternal marital status at birth of child and cubic indicators of either mother's age or years of education at birth. The demographic variables we control for include child's gender, race, ethnicity, and year of birth. We also estimate a separate set of models including average family income in early childhood (through age 5).¹⁴ Including this variable can provide an indication of the extent to which this one measure of resources can explain any marriage premium for children that we observe in the data.¹⁵ Another important reason to determine the role played by income is to understand the extent to which government transfers could potentially play a role in alleviating the outcome deficits faced by children in single-parent households (see Lerman et al. 2017).

We summarize the results of this analysis, focusing specifically on maternal age at birth of child, in **Figures 4** and **5**. Both figures consider each of our four outcomes (poverty status, high school completion, high income status, and college completion); the only difference between them is that **Figure 5** adjusts for average family income in early childhood (ages 0–5). In these figures, we plot the simulated marriage premium for children (the predicted difference in children's

¹⁴Supplemental Appendix Tables 1A and 1B report the full set of regression results.

¹⁵We note that adding these variables does not satisfy any reasonable standard of identifying a causal impact. We interpret the results from these analyses as regression-adjusted descriptive statistics.



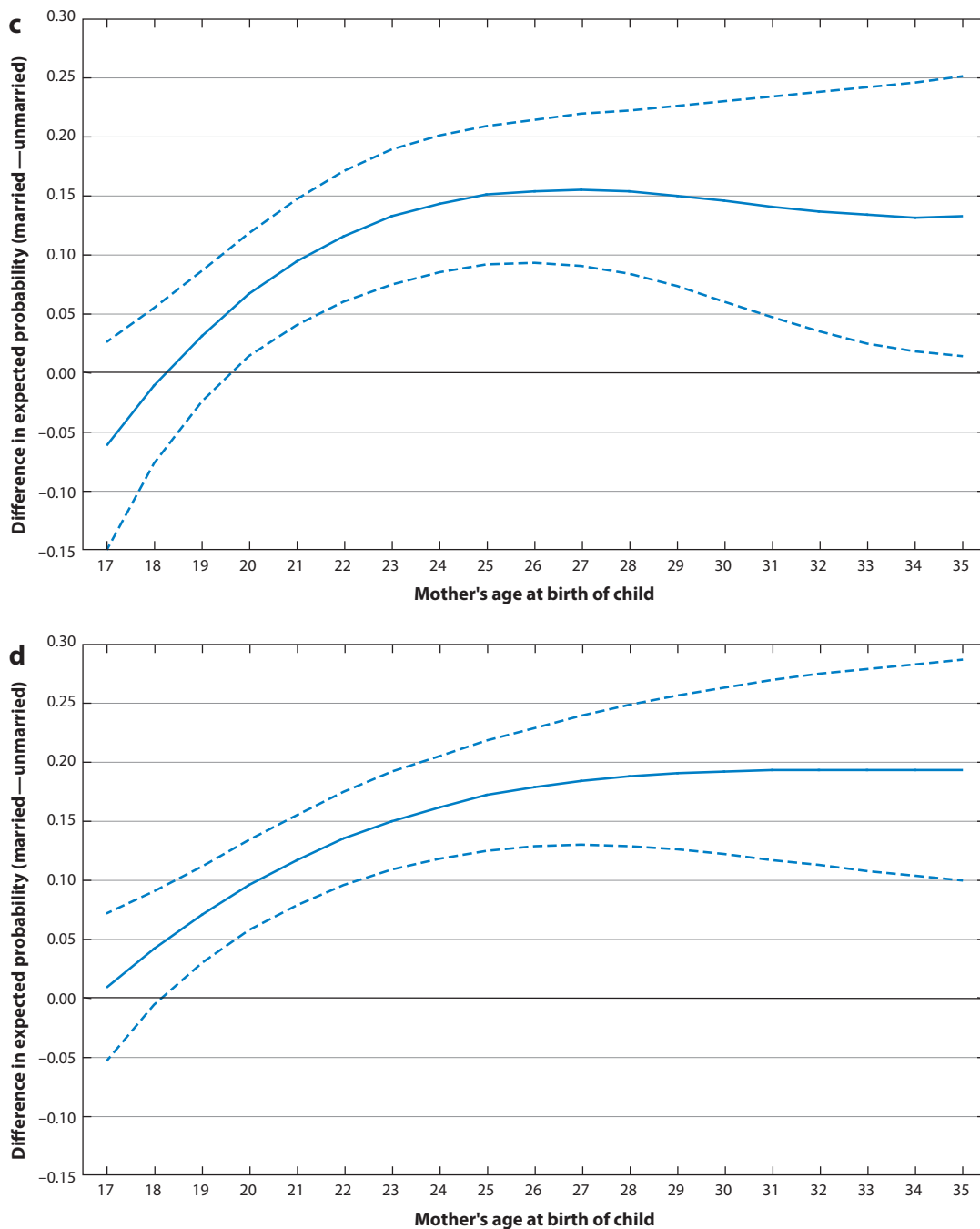
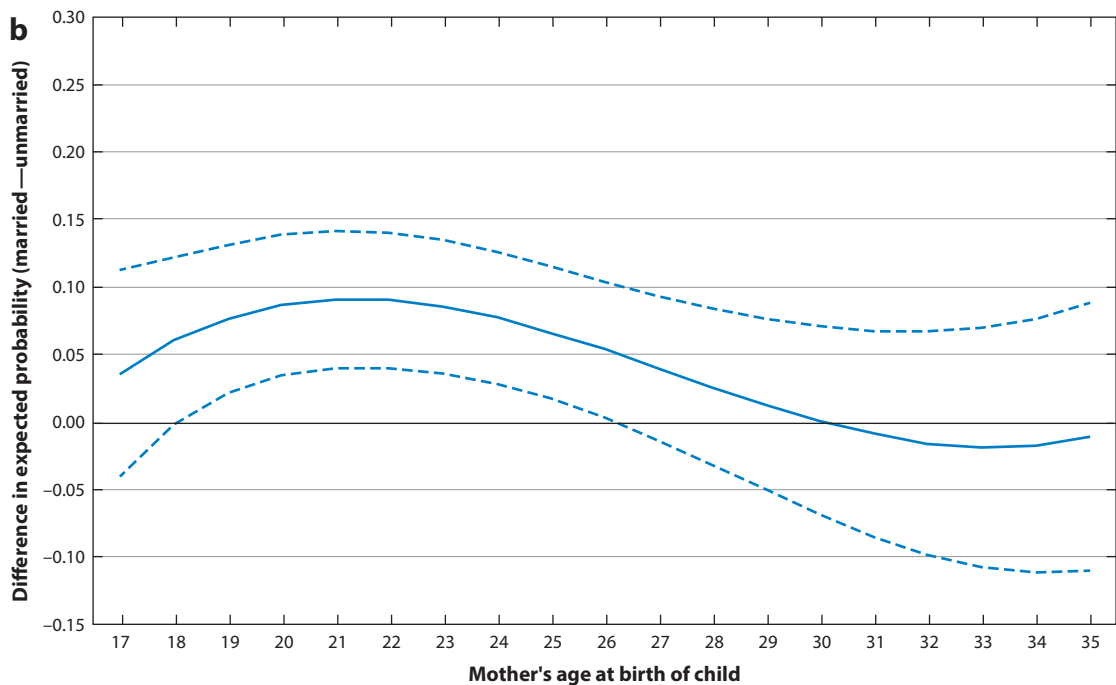
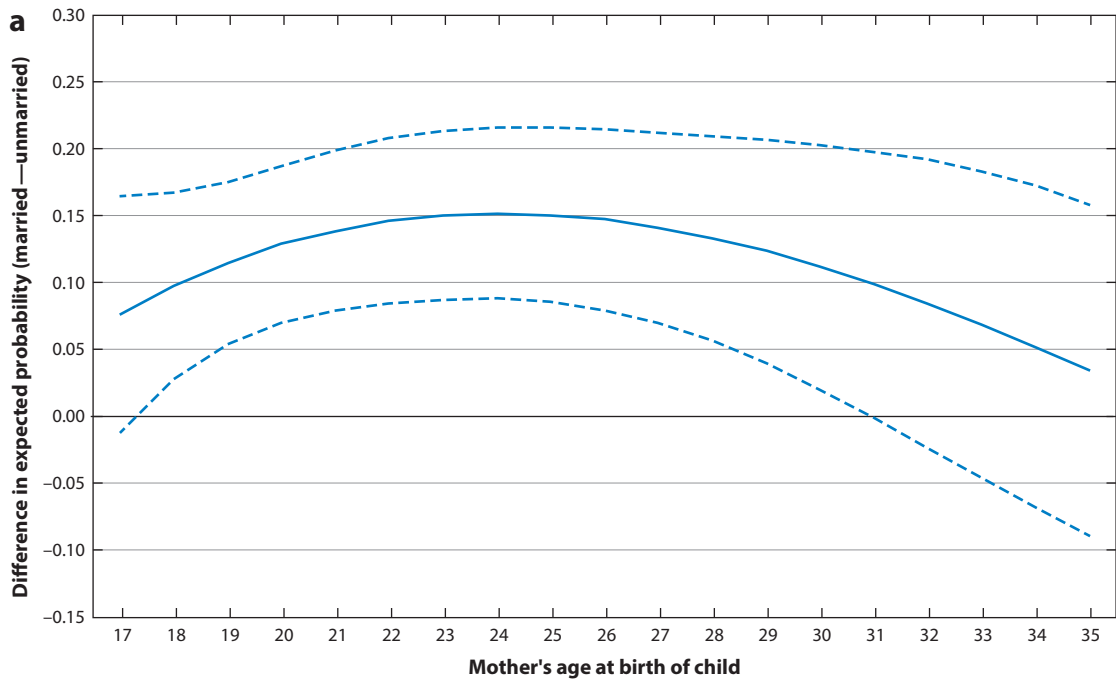


Figure 4 (Continued)

Mother's age at birth of child and the marriage premium for children (no income controls) for four outcomes: (a) being out of poverty at age 25, (b) having a high school degree by age 20, (c) being over 400% of the poverty line at age 25, and (d) having a college degree by age 25. The difference in predicted probability was obtained from regression specifications reported in **Supplemental Appendix Table 1**, controlling for gender, race, ethnicity, and year of birth. Dashed lines plot 95% confidence intervals (standard errors were obtained using the delta method). Figure created with the authors' calculations using data from the Panel Study of Income Dynamics.



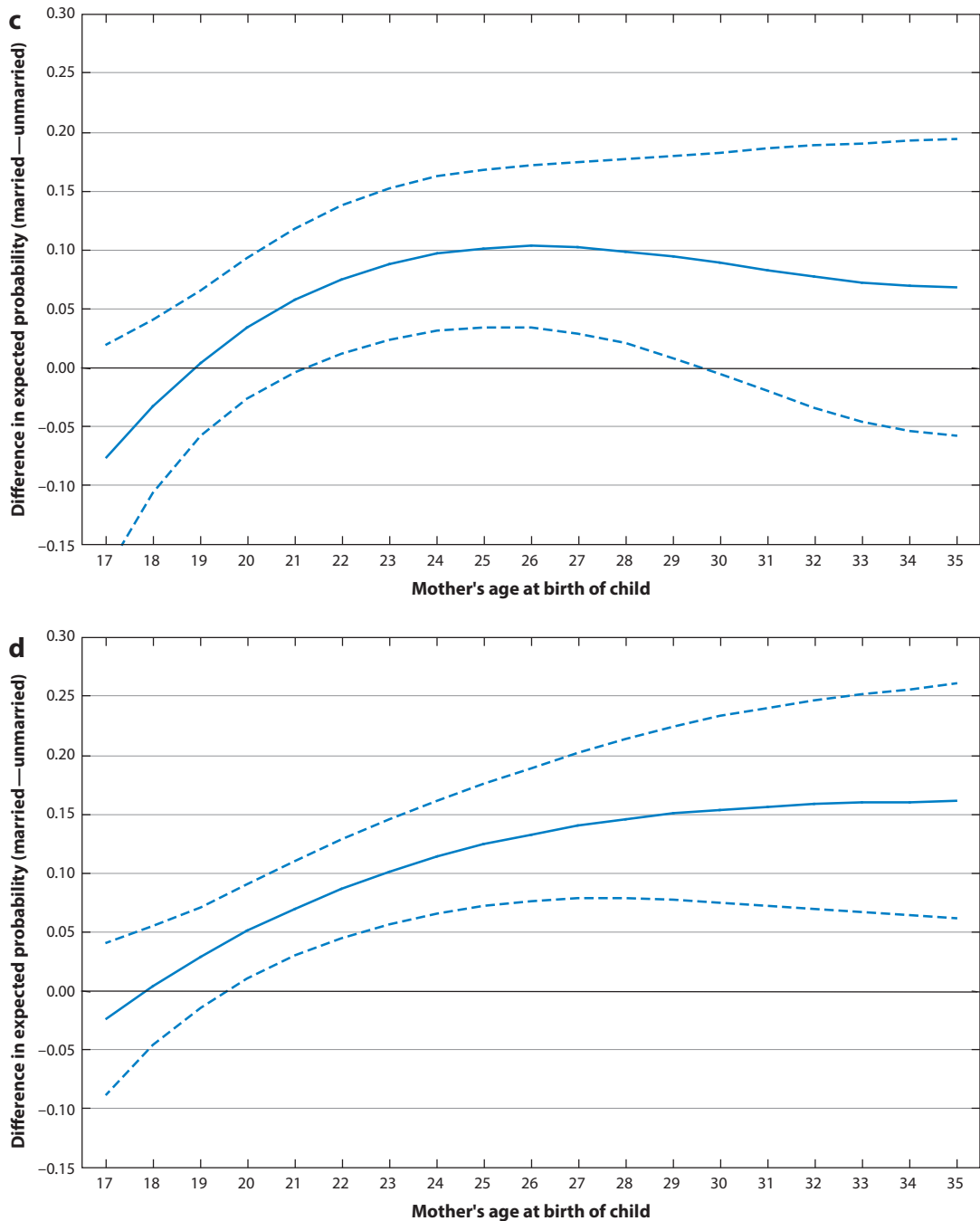


Figure 5 (Continued)

Mother's age at birth of child and the marriage premium for children (with income controls) for four outcomes: (a) being out of poverty at age 25, (b) having a high school degree by age 20, (c) being over 400% of the poverty line at age 25, (d) having a college degree by age 25. The difference in predicted probability was obtained from regression specifications reported in **Supplemental Appendix Table 2**, controlling for gender, race, ethnicity, and year of birth. Dashed lines plot 95% confidence intervals (standard errors were obtained using the delta method). Figure created with the authors' calculations using data from the Panel Study of Income Dynamics.

outcomes between those born to married and unmarried parents) as a function of their mother's age when they were born. The patterns are nonlinear because our regression model includes cubic age interactions with maternal marital status.¹⁶ The dotted lines represent 95% confidence intervals, constructed using the delta method. We repeat this analysis for mother's educational attainment; due to space constraints, those results are reported in **Supplemental Appendix Figures 1 and 2**.¹⁷

Figure 4a examines the marriage premium for children when the specific child outcome is having family income above the poverty threshold at age 25. It is consistent with the inverted U-shaped pattern that we hypothesize above for a primary outcome and that our descriptive analysis foreshadows. In fact, for the very youngest and the oldest mothers in this sample, the confidence interval for the marriage premium for children extends to near zero in terms of poverty status. The marriage premium is highest at age 24; children born to married mothers at that age are 15 percentage points more likely to be above the poverty threshold at age 25 compared to children of unmarried mothers.

Note that controlling for demographics has a large impact on the estimated marriage premium for children. In **Table 4**, those children born to married mothers between the ages of 20 and 24 were 22.6 percentage points more likely to have incomes above the poverty line at age 25. Controlling for demographics, as in **Figure 4a**, substantially reduces that gap. Doing so, though, has little impact on the inverted U-shaped pattern of the marriage premium for children. In fact, for children born to mothers under age 18 or over age 32, the estimated marriage premium in terms of children's subsequent poverty outcomes is small and not statistically significant once we control for demographics. These controls introduce something like a level shift down in the profile without altering the general shape of the profile itself.

Figure 4b displays similar patterns for high school completion. Measured in this way, the marriage premium for children peaks at maternal age 22. No statistically significant impact is estimated for maternal ages under age 18 or over age 28. The similarity in patterns between **Figure 4a** and **4b** is what we would expect from our model because the data in both panels represent primary outcomes.

Figure 4c and **4d** plot the marriage premium for the advanced outcomes of graduating from college by age 25 or having income greater than 400% of the federal poverty threshold at age 25. Consistent with the pattern reported in **Table 4**, we see that this manifestation of the marriage premium monotonically increases with mother's age until the late 20s and then asymptotes to an approximately 15– to 20–percentage point differential. The magnitude of these estimates varies little after including demographic controls. Children born to older women (presumably with greater resources) continue to benefit in the form of this advanced outcome when their mothers are married.¹⁸ We also explore heterogeneity in the marriage premium for boys and girls and find that the estimates are slightly larger for boys, but we cannot statistically distinguish between the two. For this reason, we only examine the pooled impact in the analysis below.

¹⁶We estimate these models using all available data for mothers between the ages of 14 and 45. In the figures, though, we report only simulation results of the marriage premium for children for mothers between the ages of 17 and 35. The reason for this is that there are very few marital births to mothers younger than age 17 or nonmarital births to mothers over age 35. This leads to very imprecise estimates at these ages, with standard error bands that are much wider than those between ages 17 and 35 and that always include zero.

¹⁷**Supplemental Appendix Tables A1 and A2** present the regression results that were used to generate these figures.

¹⁸If we instead use a continuous measure of log income at age 25 as the outcome of interest, we also see a monotonic increase in the observed marriage premium along the dimensions of maternal age and education.

Figure 5 repeats this analysis but controls for family income at birth as well as demographic characteristics.¹⁹ Interestingly, for the two primary outcomes (poverty status and high school completion), the premium to children associated with marriage does not appear to be accounted for by a simple control for average family income in early childhood. The patterns and levels displayed in **Figures 5a** and **5b** are very similar to those reported in **Figures 4a** and **4b**, respectively. The finding that controlling for family income in early childhood does not appreciably change the estimated marriage premium might indicate that other resources associated with marriage are driving the observed premium. Such resources might include greater family stability, more total parental time, and lower levels of maternal stress. It could also be that average family income during early childhood is not a sufficient proxy for family income over a child's life, and a more comprehensive measure of income would indicate that family income is indeed an important factor in explaining the marriage premium.

In terms of the advanced outcomes, average family income during early childhood seems to be an important contributing factor in determining college completion, but it does not have much of an impact on the estimated premium for having a high income at age 25. Although the pattern in **Figure 5c** (for college completion) is very similar to that in **Figure 4c**, the marriage premium to children asymptotes to a lower value, 15 percentage points rather than 20, when adjusted for family income.

5. DISCUSSION

Numerous empirical studies written across social science disciplines indicate that children raised in married-biological-parent households tend to do better on a host of outcomes, both during childhood and into adulthood, compared to children raised by an unmarried mother. Given that marital status is not randomly assigned to an individual, a precise causal estimate of marriage per se remains elusive. Nonetheless, as we noted at the outset, the strength of the association and its robustness to a myriad of controls is at least strongly suggestive of a causal relationship. In this review, we label this observed difference the marriage premium for children. When viewed through the lens of economics, it becomes clear that such a premium could be both explained and anticipated by an economic consideration of potential resources.

The academic literature and policy conversation about nonmarital childbearing have generally not distinguished among different categories of nonmarital childbearing. We propose a framework for evaluating nonmarital childbearing and children's outcomes that has clear heterogeneous predictions. Specifically, we propose that the marriage premium in terms of children's outcomes will depend on a mother's own level of resources, the nature of marriage markets and the additional resources a spouse would bring to the household, and the relevant production function for a specific outcome. Using data from the PSID, we find that the predictions of our framework are consistent with the patterns in the data.

We divide children's later-life outcomes into those that are primary (high school graduation and not living in poverty) and those that are advanced (college graduation and having a high income). For primary outcomes, the data indicate the largest marriage premiums for children of women who give birth in their early to mid-20s and for children of high school graduates. The relative marriage advantage is lower for children of mothers who are in their teens or in their 30s, as well as for children of high school dropouts or college graduates. We interpret these patterns as being consistent

¹⁹Following Lopoo & Deleire (2014) and Lerman et al. (2017), we control for income by dividing our measure of income by the square root of the household size to account for economies of scale.

with the prediction that the incremental resources associated with marriage are too small to have much of an impact for young and less educated mothers, and are not necessary for older and more educated mothers, to substantially increase the likelihood of obtaining a primary outcome. Interestingly, for advanced outcomes, the marriage premium monotonically increases with observed maternal age and education. This suggests that, even for mothers with a high level of resources of their own, the additional resources brought into a married household have a sizeable impact on the likelihood that a child graduates from college and achieves a high level of adult income.

The conceptual framework and descriptive empirical evidence of this paper imply that it is important to distinguish across maternal circumstances when evaluating the issue of nonmarital childbearing and children's outcomes. They also have implications for thinking about recent trends toward higher rates of nonmarital fertility among older and more educated women. In previous decades, nonmarital childbearing was quite uncommon except among women in their teens and among those without a high school degree. However, in recent decades, we have witnessed a dramatic rise in nonmarital childbearing among women with a high school degree and women in their 20s. Our results from PSID data on cohorts born during the 1960s, 1970s, and 1980s suggest that this is precisely where the marriage premium is largest for children.

The extent to which these results hold lessons for predicting heterogeneity in the marriage premium for children being born to unmarried mothers today will depend on how maternal resources, marriage markets, paternal resources, and the production function of child outcomes have changed. The key takeaway from our framework and empirical results is that the marriage premium depends crucially on resource context. We have focused on the economic resources that fathers would bring to marriage. This is related to a leading explanation for the high rates of nonmarital childbearing among less educated and minority women: the notion that there is a dearth of marriageable men, generally understood to mean men who would bring economic stability, in certain segments of the population (Wilson & Neckerman 1986, Wilson 1987). In recent decades, the economic position of the median male in terms of education and earnings has declined relative to both earlier cohorts (Greenstone & Looney 2011) and contemporaneous women (Autor & Wasserman 2013). The declines in employment and earnings have been especially pronounced among men with high school degrees (Autor & Wasserman 2013, Kearney et al. 2015). If part of the reason that rates of nonmarital childbearing have increased among women with high school degrees is that there has been an erosion in the economic prospects of the men with whom they partner,²⁰ then the marriage premium their children would enjoy is likely to be lower than the premiums we have found among earlier cohorts.

As this discussion highlights, the economic causes of nonmarital childbearing are likely related to the resulting economic consequences of nonmarital childbearing. Empirically separating out these causes and effects is a challenging task. Much more research into the complex issue of nonmarital childbearing and what it implies for the economic well-being of affected children is needed. The framework and empirical results we have put forth in this review demonstrate the importance of focusing on heterogeneous effects related to potential household resources.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

²⁰A new study by Cherlin et al. (2016) provides evidence supporting this hypothesis from the 1997 cohort of the National Longitudinal Survey of Youth.

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